

assumption that the economy has ample excess capacity and unemployed resources, so that output can be demand-determined.

2.3.5 Limitations of the Keynesian Model

In the beginning of this section we mentioned that the Keynesian model is more relevant for an economy facing recessionary conditions, having excess production capacity in place. In economies having supply constraints (i.e., where there are supply bottlenecks), an increase in demand is likely to result in a rise in prices rather than in output. Second, Keynesian model would not apply to economies that are operating near full employment, where increase in output would require increase in factor and goods prices. Third, Keynesian model is generally used for macroeconomic analysis pertaining to the 'short run'. Normally, once wage contracts are written they last for some time; also, prices printed on sales catalogues are expected to be valid for some time. So the assumption of wage and price rigidity is quite reasonable for the short run.

Check Your Progress 2

1. State the assumptions on which the Keynesian model is based.

.....

.....

.....

.....

2. You are given the following information for a closed economy: $C = 10 + 0.75 Y$ and $\bar{I} = 20$.

- a) Find the aggregate demand function and calculate equilibrium income for this economy.
- b) Suppose autonomous investment increases to 25. How would equilibrium income change?
- c) What is the value of the multiplier?
- d) How would the value of the multiplier in (b) change if we introduce the government sector which imposes (i) only a lump sum tax, and (ii) only a proportional tax where $t = 0.1$?
- e) How would the value of the multiplier in (b) change if the economy is open and $MPM = 0.3$?

.....

.....

.....

.....

3. By using appropriate diagram of the Keynesian model, explain how situations of disequilibrium (excess demand or excess supply) are self-correcting.

.....

.....

.....

.....

2.4 AN ALTERNATIVE VIEW OF EQUILIBRIUM

The Keynesian model can alternatively be described through the saving-investment relationship. When saving is equal to planned investment, the economy is operating at the equilibrium level of output.

2.4.1 Saving Function

The saving function is a mirror image of the consumption function that gives the relationship between planned saving (S) and income (Y) in the economy. In the two-sector model, with only households and firms, aggregate income is either consumed or saved. Thus,

$$Y = C + S \quad \dots (2.11)$$

The decision to consume more is essentially the same as the decision to save less.

The saving function may be derived as follows:

$$S = Y - (a + bY) = -a + (1-b)Y \quad \dots (2.12).$$

Note that the intercept of the saving function is negative, indicating that ‘dis-saving’ (running down past saving) occur at relatively low levels of income. The slope of the saving function is $(1-b)$, which is known as the *marginal propensity to save* (MPS). Note that $MPS = (1-MPC)$, so that higher the MPC, lower is the MPS. In economic analysis it is often assumed that richer households have higher MPS compared poor households.

In Fig. 2.6 we derive the saving function diagrammatically. The top panel represents the consumption function C, while the lower panel represents the saving function.

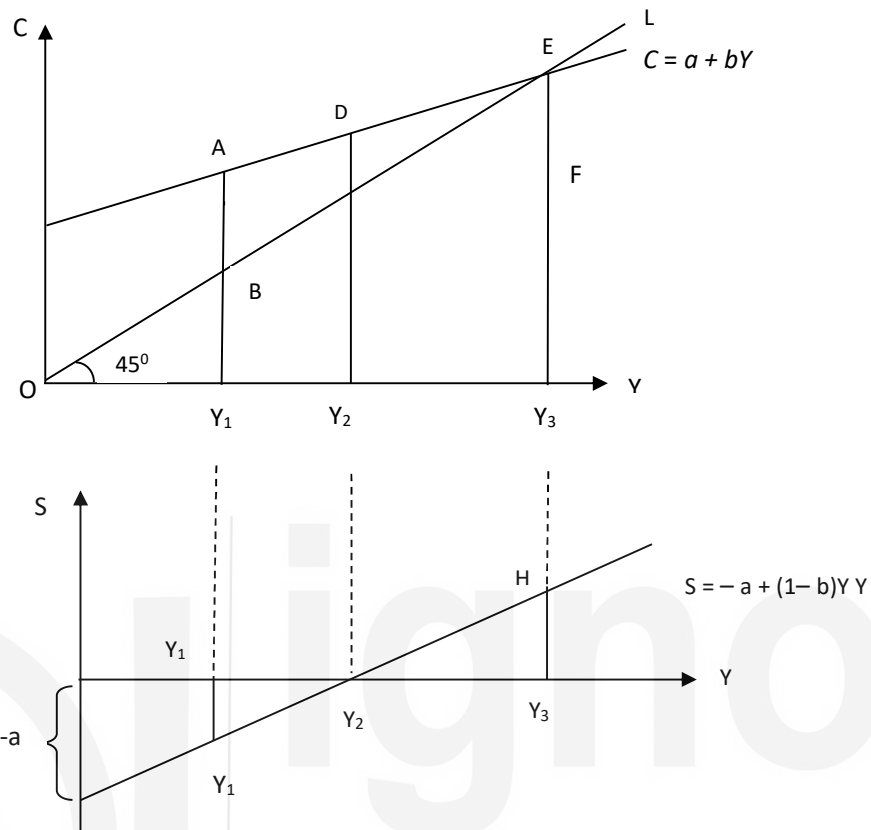


Fig. 2.6: Saving Function

At income level Y_2 , where C intersects OL (the 45° line), planned consumption DY_2 is equal to income OY_2 ($= DY_2$) and planned saving is zero. At this level of income, the saving function intersects the x-axis. For all income levels less than OY_2 planned consumption exceeds income so that saving is negative. For example, in the top panel, at income level OY_1 , planned consumption AY_1 is greater than income BY_1 ($= OY_1$) and the difference is reflected in negative saving Y_1G in the lower panel. For income levels such as OY_3 , which is higher than OY_2 , planned saving is positive (HY_3 , in the lower panel).

2.4.2 Determination of Equilibrium Output

In a two-sector economy with only households and firms, equilibrium in the output market occurs at the level of income (or output) at which planned saving is equal to planned investment. This can be shown as follows: Suppose investment is autonomous and the investment function is given by $I = \bar{I}$ as in (2.2a). Recall that $Y = C + \bar{I}$ at equilibrium. We know that income is either consumed or saved, i.e., $Y = C + S$. Thus, we can put these together to write

$$C + S = Y = C + \bar{I}, \text{ or, } S = \bar{I} \quad \dots (2.13)$$

In a closed economy without government, at equilibrium, planned saving must be equal to planned investment. Abstaining from current consumption (i.e., saving)